

AQA A-Level Further Mathematics (7367)

Question Paper

Continuous Random Variables

SD5 - Functions of a CRV

Name: _____

Class: _____

Date: _____

Time: 57 min.

Marks: 47 marks

Comments:

Q1.

The continuous random variable X has a cumulative distribution function defined by

$$F(x) = \begin{cases} 0 & x < -5 \\ \frac{x+5}{20} & -5 \leq x \leq 15 \\ 1 & x > 15 \end{cases}$$

- (a) Show that, for $-5 \leq x \leq 15$, the probability density function, $f(x)$, of X is given by $f(x) = \frac{1}{20}$.

(1)

- (b) Find:

(i) $P(X \geq 7)$;

(1)

(ii) $P(X \neq 7)$;

(1)

(iii) $E(X)$;

(1)

(iv) $E(3X^2)$.

(3)

(Total 7 marks)

Q2.

The continuous random variable X has probability density function given by

$$f(x) = \begin{cases} \frac{1}{2}(x^2 + 1) & 0 \leq x \leq 1 \\ (x-2)^2 & 1 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f .

(3)

- (b) Calculate $P(X \leq 1)$.

(3)

- (c) Show that $E(X^2) = \frac{4}{5}$.

(5)



- (d) (i) Given that $E(X) = \frac{19}{24}$ and that $\text{Var}(X) = \frac{499}{k}$, find the numerical value of k . (3)
- (ii) Find $E(5X^2 + 24X - 3)$. (2)
- (iii) Find $\text{Var}(12X - 5)$. (2)
- (Total 18 marks)

Q3.

The continuous random variable X has the probability density function given by

$$f(x) = \begin{cases} 3x^2 & 0 < x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Determine:

(i) $E\left(\frac{1}{X}\right)$; (3)

(ii) $\text{Var}\left(\frac{1}{X}\right)$. (4)

- (b) Hence, or otherwise, find the mean and the variance of $\left(\frac{5 + 2X}{X}\right)$. (5)
- (Total 12 marks)

Q4.

The discrete random variable X has

$$E(X) = 4 \quad \text{and} \quad E(X^2) = 17.2.$$

- (a) Find the value of $\text{Var}(X)$. (2)
- (b) A circle has radius $(X + 8)$.

Find, in terms of π , values for the mean and variance of the **circumference**, C , of the circle. (4)

- (c) The **area**, S , of the circle, with radius $(X + 8)$, is to be expressed in the form

$$\pi(X^2 + aX + b),$$

where a and b are positive constants.

(i) Find values for a and b .

(2)

(ii) Hence find, in terms of π , the value of $E(S)$.

(2)

(Total 10 marks)

